COMPUTER ENGINEERING
QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S2
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Seatwork 7.2 **Algorithm Analysis: Best-Case and Worst-Case Time Complexity**

1. PROCEDURE OUTPUT

Instructions: Analyze each algorithm carefully. For every item, determine the following:

1. **Best-Case Time Complexity**
2. **Worst-Case Time Complexity**
3. **Brief explanation** of how you arrived at your answers.

Use asymptotic notation such as **O(1), O(log n), O(n), O(n log n), O(n^2)**, etc.

1. Finding an Element

```C/C++
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        return i;
    }
}
```

Best-Case Time Complexity	O(1)
Worst-Case Time Complexity	O(n)
explanation	In the best case, target is at `arr[i]` so one comparison is needed while the worst case target is at the last position, so the loop checks all n elements.

2. Printing All Elements

```C/C++
for (int i = 0; i < n; i++)
```

```
{
    cout << arr[i] << endl;
}
```

Best-Case Time Complexity	O(n)
Worst-Case Time Complexity	O(n)
explanation	There are no conditional break so the loop always run regardless of data.

3. Nested Loops

```C/C++
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i << " " << j << endl;
    }
}
```

Best-Case Time Complexity	$O(n^2)$
Worst-Case Time Complexity	$O(n^2)$
explanation	The outer loop runs at n times then for every outer iteration the inner loop also runs n times which means n\times n = n^2

4. Conditional Processing

```C/C++
if (n > 100)
{
    for (int i = 0; i < n; i++)
    {
        cout << arr[i];
```

```
    }
}
else
{
    cout << "Small input";
}
```

Best-Case Time Complexity	O(1)
Worst-Case Time Complexity	O(n)
explanation	If n is 100 or less, it only prints once. If n is greater than 100, the loop runs n times.

5. Fixed Inner Loop

C/C++
```
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < 10; j++)
    {
        cout << i << j;
    }
}
```

Best-Case Time Complexity	O(n)
Worst-Case Time Complexity	O(n)
explanation	The outside loop runs n times, while the inside loop only runs 10 times. So it is still O(n).

6. Two Consecutive Loops

C/C++
```
for (int i = 0; i < n; i++)
```

```
{
    cout << arr[i];
}

for (int j = 0; j < n; j++)
{
    cout << arr[j] * 2;
}
```

Best-Case Time Complexity	O(n)
Worst-Case Time Complexity	O(n)
explanation	The first loop runs n times and the second loop runs n times one after another. This results in O(n) + O(n) = O(2n), which simplifies to O(n).

7. Dividing by Two

C/C++
```
int x = n;

while (x > 1)
{
    x = x / 2;
}
```

Best-Case Time Complexity	O(log n)
Worst-Case Time Complexity	O(log n)
explanation	Every iteration divide x by 2 which means: n n/2 n/4 n/8 n/

8. Nested Loop With Different Limits

```
C/C++
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < i; j++)
    {
        cout << i << j;
    }
}
```

Best-Case Time Complexity	$O(n^2)$
Worst-Case Time Complexity	$O(n^2)$
explanation	The inner loop gets bigger as i increases. Then, the number of operations is around n^2.

9. Early Termination

```
C/C++
for (int i = 0; i < n; i++)
{
    if (arr[i] == target)
    {
        cout << "Found";
        break;
    }
}
```

Best-Case Time Complexity	O(1)
Worst-Case Time Complexity	O(n)
explanation	If the target is first, the loop stops immediately. If it is last or missing, it checks everything.

10. Two-Level Search

```C/C++
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        if (arr[i] == target)
        {
            break;
        }
    }
}
```

Best-Case Time Complexity	O(n)
Worst-Case Time Complexity	$O(n^2)$
explanation	In the best case, the inner loop stops early, but the outside loop still continues. In the worst case, both loops run completely.

11. Linear Loop With Logarithmic Inner Work

```C/C++
for (int i = 0; i < n; i++)
{
    int x = n;

    while (x > 1)
    {
        x = x / 2;
    }
}
```

Best-Case Time Complexity	O(n log n)
Worst-Case Time Complexity	O(n log n)
explanation	

12. Conditional Nested Loop

```C/C++
if (arr[0] == target)
{
    cout << "Found";
}
else
{
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            cout << arr[i] + arr[j];
        }
    }
}
```

Best-Case Time Complexity	O(1)
Worst-Case Time Complexity	$O(n^2)$
explanation	If the first condition is true, it only prints once. If not , it runs the two loops, giving $O(n^2)$.

13. Finding the Maximum

```C/C++
int max = arr[0];

for (int i = 1; i < n; i++)
{
    if (arr[i] > max)
    {
        max = arr[i];
    }
}
```

Best-Case Time Complexity	O(n)

Worst-Case Time Complexity	O(n)
explanation	

14. Three Nested Loops

```C/C++
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        for (int k = 0; k < n; k++)
        {
            cout << i + j + k;
        }
    }
}
```

Best-Case Time Complexity	$O(n^3)$
Worst-Case Time Complexity	$O(n^3)$
explanation	

15. Comprehensive Analysis

```C/C++
for (int i = 0; i < n; i++)
{
    for (int j = 0; j < n; j++)
    {
        cout << i + j + k;
        }
    }
}
```

Best-Case Time Complexity	

Worst-Case Time Complexity	
explanation	

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

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